

Portfolio Question 4.2.P - Mixture Modelling

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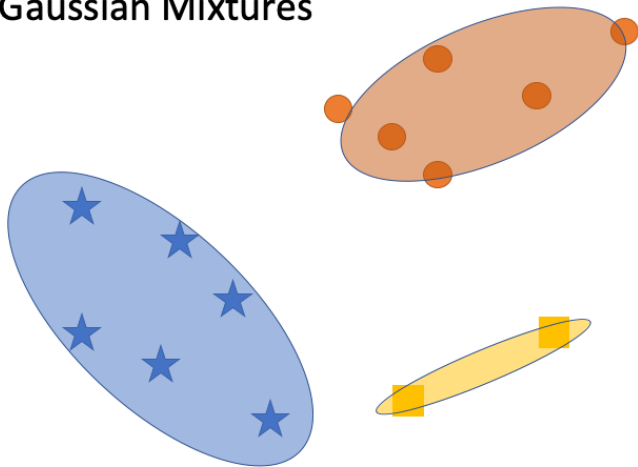
Portfolio 4.2.P (v4.0.0)

Portfolio Question 4.2.P - Mixture Modelling

- ▶ “Gaussian Mixture Modelling” theoretically dominates “soft K-means”.
 - ▶ Explain Gaussian Mixture modelling and how it relates to K-means as described in the lecture.
 - ▶ Is this claim true? If so, why would we ever use K-means?

Gaussian Mixture Modelling

Gaussian Mixtures



Gaussian Mixture Modelling

- ▶ Randomly **initialise** K locations as initial cluster means μ_k , each with an initial covariance Σ_k (can just be spherical)
- ▶ Iteratively, until convergence:
 1. Compute the **density of each cluster** at each point
$$d_{ik} = K_k(x_i | \mu_k, \Sigma_k)$$
 2. Compute the **probability** (“soft”) of each cluster for each point:
$$p_{ik} = d_{ik} / \sum_{k'} d_{ik'}$$
 3. **Update the cluster** parameters accounting for the probabilistic memberships
- ▶ In practice, we still want to:
 - ▶ Use several starting values
 - ▶ Use “intelligent” initial guesses
- ▶ This “Expectation Maximization” approach speeds convergence over “hard” K-means
- ▶ **Computational complexity** is $O(N^2)$, though the constant is larger than for K-means. What is the dependency on K ?

Expectation Maximization

- ▶ **Expectation-Maximization (EM)** is an optimization tool for problems with a latent parameter $\mathbf{Z}$ of the form:

$$L(\theta, \mathbf{X}) = p(\mathbf{X}|\theta) = \int p(\mathbf{X}, \mathbf{Z}|\theta) d\mathbf{Z}$$

- ▶ Where we wish to maximise the Likelihood $L(\theta, \mathbf{X})$ with respect to θ , marginalising out $\mathbf{Z}$.
- ▶ In soft K-means, $\mathbf{Z}$ is the probability of belonging to each cluster; θ is the location of the clusters.
- ▶ EM solves this by iteratively:
 - ▶ Computing the **Expected value** of the latent $\mathbb{E}(\mathbf{Z}|\theta)$,
 - ▶ Computing the **Maximum likelihood** estimate $p(\mathbf{X}, \mathbf{Z}|\theta)$.
- ▶ EM provably always improves $L(\theta, \mathbf{X})$.

Gaussian Mixture Modelling

- ▶ GMMs work very well on a range of problems.
- ▶ However, choosing Σ and K can be awkward
- ▶ One solution is to use a (semi)Bayesian paradigm:
 - ▶ Fit the clusters using EM as in regular GMMs
 - ▶ Use Bayesian Model selection (BIC) to choose a model for Σ and select K
 - ▶ Σ choices: ellipsoid vs circular, volume, shape, orientation
 - ▶ Changes the dimension of Σ , hence affects BIC
- ▶ This isn't reliable **model selection** for whether GMM is appropriate, but it is good selection for what shape Σ to use
- ▶ In R: `library(mclust)`

Example: K-means clustering

- ▶ Run K-means clustering on the whole example dataset:

```
km.all.raw=lapply(1:10,function(i){  
  km=kmeans(testdata_all_scaled,centers=i,nstart=10)  
})
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Example: K-means clustering

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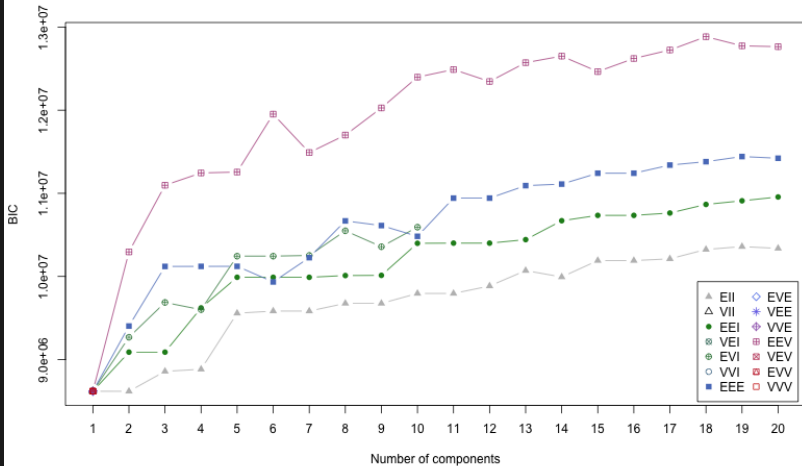
- ▶ **Spectral clustering** just means running the same clustering algorithm on the top PCs in a PCA/SVD

```
km.all.svd=lapply(1:10,function(i){  
  km=kmeans(testdata_all.svd$u,centers=i,nstart=10)  
})
```

Example: GMM using mclust

```
library("mclust")
mc.all=mclustBIC(testdata_all.svd$u,G=1:20)
# mclustBIC Compares lots of models
mc.assignments=lapply(1:20,function(i){
  tmp=mclustModel(testdata_all.svd$u,mc.all,G=i)
  apply(tmp$z,1,which.max)
}) # extract the results for the best models
```

Example: GMM using mclust: diagnostics



Discussion

- ▶ K-means is a special case of GMM when all the clusters are spheres of the same size
 - ▶ It therefore “dominates”
 - ▶ But we have to estimate it
- ▶ GMM vs K-means:
 - ▶ many more parameters: the shape of each cluster, etc, which must be inferred
 - ▶ more computationally costly
- ▶ Does the algorithm converge to a good solution?
- ▶ How much data do we need?
- ▶ Is it interpretable?